

A Disproof of Conjecture 00000001238

Log-concavity of Jacobian invariant factors, and the equality class

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Abstract

Conjecture 00000001238 asserts that (i) the invariant-factor sequence of the Jacobian (sand-pile group) of every graph is log-concave, and (ii) equality in the concavity holds *exactly* for direct sums of cycles. Both clauses are false. The complete bipartite graph $K_{2,4}$ has Jacobian $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/8$, whose invariant-factor sequence $(2, 2, 8)$ violates the middle concavity inequality ($2 \cdot 2 = 4 < 2 \cdot 8 = 16$); and K_4 has Jacobian $\mathbb{Z}/4 \times \mathbb{Z}/4$, whose sequence $(4, 4)$ *has* equality although K_4 is not a direct sum of cycles (the edge $\{0, 1\}$ lies on the two triangles $\{0, 1, 2\}$ and $\{0, 1, 3\}$). Moreover the two clauses are jointly unsatisfiable even on the class the conjecture nominates: the cactus $C_3 \vee C_3 \vee C_6$ is a direct sum of cycles yet has invariant factors $(3, 3, 6)$, which are not log-concave ($9 < 18$). An exhaustive sweep over all 2^{15} graphs on six labeled vertices (26,704 connected) corroborates the picture. All numeric facts are certified in core Lean 4 with zero axioms and no `sorry`, and reproduced by a standalone script.

1 The conjecture

“Definition: The invariant-factor sequence of the Jacobi group encodes the torsion structure of the graph Laplacian. Conjecture: Log-concavity of the invariant-factor sequence of the torsion group holds for all graphs; and equality in the concavity holds exactly for direct sums of cycles. (Jacobian invariant factors log-concave)”

2 Conventions and reading

For a finite connected graph G with vertices V , the *Jacobian* $\text{Jac}(G)$ is the cokernel $\mathbb{Z}^V / \text{rows}(L)$ of the graph Laplacian L ; equivalently $\mathbb{Z}^{|V|-1} / \text{rows}(L^\circ)$ for any reduced Laplacian L° (delete one row and the matching column). Its *invariant-factor sequence* is the nontrivial part $d_1 \mid d_2 \mid \cdots \mid d_k$ ($d_i > 1$) of the Smith normal form diagonal of L° ; this is the reading fixed by the pipeline’s own confirmed instance (SNF $(1, 1, 2, 2, 8)$, nontrivial factors $[2, 2, 8]$, $2^2 = 4 < 2 \cdot 8 = 16$). *Log-concavity* is the system of interior inequalities $d_i^2 \geq d_{i-1}d_{i+1}$ for $1 < i < k$; *equality* means every interior inequality is an equality. A *direct sum of cycles* is a graph each of whose blocks is a single edge or a cycle (a cactus: cycles joined at cut vertices), whose Jacobian is then the direct sum of the $\mathbb{Z}/(\text{cycle length})$.

Remark 1 (Convention robustness). Our counterexamples do not depend on these conventions. The failure of $K_{2,4}$ occurs at an *interior* triple $(2, 2, 8)$ with a strict inequality, so it persists under the full-sequence reading (with leading 1s), under endpoint augmentation, and under reversal. The failure of the “exactly” clause is witnessed by K_4 (sequence $(4, 4)$, equality $4 \cdot 4 = 4 \cdot 4$) and, with interior equalities, by K_5 (sequence $(5, 5, 5)$, all equalities $25 = 25$, $|\text{Jac}(K_5)| = 5^3 = 125$); neither

is a cactus under any reading. If instead “direct sum of cycles” were read group-theoretically, the clause would be vacuous (every finite abelian group is a direct sum of cyclic groups), so the graph-theoretic reading is the only substantive one.

3 Counterexample A: log-concavity fails at $K_{2,4}$

Take parts $A = \{0, 1\}$, $B = \{2, 3, 4, 5\}$. Deleting vertex 0 leaves the reduced Laplacian

$$L^\circ(K_{2,4}) = L_{24} = \begin{pmatrix} 4 & -1 & -1 & -1 & -1 \\ -1 & 2 & 0 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 \\ -1 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

The following integer matrices are unimodular, $\det U_{24} = \det V_{24} = -1$ (each is a product of elementary row, resp. column, operations applied to L_{24} ; all entries are verified by the kernel):

$$U_{24} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 7 & 0 & 0 \\ 2 & 1 & 6 & 1 & 0 \\ 2 & 1 & 5 & 1 & 1 \end{pmatrix}, \quad V_{24} = \begin{pmatrix} 0 & -1 & 0 & 0 & 2 \\ -1 & -4 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & -1 & 5 \end{pmatrix},$$

$$U_{24} L_{24} V_{24} = \text{diag}(1, 1, 2, 2, 8).$$

Hence

$$\text{Jac}(K_{2,4}) \cong \mathbb{Z}/1 \times \mathbb{Z}/1 \times \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/8 \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/8,$$

with invariant-factor sequence $(2, 2, 8)$. As a sanity check, the order equals the number of spanning trees, $|\text{Jac}(K_{2,4})| = 1 \cdot 1 \cdot 2 \cdot 2 \cdot 8 = 32 = 2^{4-1} \cdot 4^{2-1}$, which is the matrix-tree value $\tau(K_{m,n}) = m^{n-1}n^{m-1}$ at $(m, n) = (2, 4)$. The middle concavity condition for $(2, 2, 8)$ requires $d_2^2 \geq d_1 d_3$, i.e.

$$2 \cdot 2 = 4 \not\geq 2 \cdot 8 = 16,$$

which fails.

Proposition 1. *The invariant-factor sequence of $\text{Jac}(K_{2,4})$ is $(2, 2, 8)$, which is not log-concave. Clause (i) of Conjecture 00000001238 is false.*

4 Counterexample B: equality without being a cycle sum

4.1 K_4

The reduced Laplacian of K_4 (vertices $\{0, 1, 2, 3\}$, vertex 0 deleted) is $L^\circ(K_4) = \text{diag-off}(3, -1, -1; -1, 3, -1; -1, -1, 3)$ and

$$U_4 = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}, \quad V_4 = \begin{pmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 2 \end{pmatrix}, \quad U_4 L^\circ(K_4) V_4 = \text{diag}(1, 4, 4),$$

with $\det U_4 = \det V_4 = 1$. So $\text{Jac}(K_4) \cong \mathbb{Z}/4 \times \mathbb{Z}/4$ with sequence $(4, 4)$: the (length-two) concavity condition $d_1^2 \geq d_1 d_2$ reads $16 \geq 16$ and holds *with equality*. But K_4 is not a direct sum of cycles:

- the edge $\{0, 1\}$ lies on the two distinct triangles $\{0, 1, 2\}$ and $\{0, 1, 3\}$, while every edge of a direct sum of cycles lies on at most one cycle;
- equivalently, an exhaustive check of all 2^6 graphs on four vertices shows every connected cactus has at most 4 edges and at most $\tau \in \{1, 3, 4\}$ spanning trees (there are exactly 31 of them), whereas K_4 has 6 edges and $\tau(K_4) = 4^{4-2} = 16$.

So the “only if” direction of clause (ii) fails. The same failure in a form with interior equalities is given by K_5 : $\text{diag}(1, 5, 5, 5)$, sequence $(5, 5, 5)$, all interior equalities $25 = 25$, $|\text{Jac}(K_5)| = 125 = 5^3$, and K_5 is very far from a cactus. (On six vertices, K_6 itself has sequence $(6, 6, 6, 6)$ with all interior equalities $36 = 36$.)

Proposition 2. $\text{Jac}(K_4) \cong \mathbb{Z}/4 \times \mathbb{Z}/4$ has equality in the concavity, but K_4 is not a direct sum of cycles. Clause (ii) of Conjecture 00000001238 is false.

5 The two clauses are jointly unsatisfiable

The conjecture’s own equality class does not even satisfy clause (i). The cactus $C_3 \vee C_3 \vee C_6$ (three cycles joined at one vertex; a direct sum of cycles in the conjecture’s sense) has Jacobian $\mathbb{Z}/3 \times \mathbb{Z}/3 \times \mathbb{Z}/6$ with sequence $(3, 3, 6)$, and

$$3 \cdot 3 = 9 \not\geq 3 \cdot 6 = 18.$$

Thus no interpretation makes both clauses true: any graph class witnessing “exactly” must contain cacti, and cacti already break log-concavity. (A caution for the reading: the wedge $C_3 \vee C_4$ has Jacobian $\mathbb{Z}/3 \times \mathbb{Z}/4 \cong \mathbb{Z}/12$ by the Chinese remainder theorem, since $\gcd(3, 4) = 1$; its invariant-factor sequence is the *single* factor (12) . Divisibility of the cycle lengths controls when a cactus Jacobian stays decomposed.)

6 Exhaustive corroboration on six vertices

Sweeping all 2^{15} graphs on six labeled vertices and computing the Smith normal form of each reduced Laplacian by determinantal divisors (gcds of all $k \times k$ minors):

connected graphs on 6 labeled vertices	26,704
nontrivial sequences of length ≥ 2 that violate an interior inequality	75
graphs with torsion exactly $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/8$	15 ($K_{2,4}$ among them)
equality sequences of length ≥ 2 on non-cacti	3,793
of which of length ≥ 3 (e.g. K_6 , $(6, 6, 6, 6)$)	31

So both failure modes are ubiquitous already on six vertices.

7 Machine verification

The directory `lean4/` contains a core-only Lean 4 project (toolchain `leanprover/lean4:v4.33.1`) certifying, with **zero axioms and no sorry** (`Check.lean` audits all eleven theorems; none depends on any axiom):

- `Tlmc1238.snf_L24`: $U_{24}L_{24}V_{24} = \text{diag}(1, 1, 2, 2, 8)$, $\det U_{24} = \det V_{24} = -1$;

- `Tlmc1238.snf_LK4`: $U_4 L^\circ(K_4) V_4 = \text{diag}(1, 4, 4)$, $\det U_4 = \det V_4 = 1$;
- `Tlmc1238.lap24_is_minor`, `lapK4_is_minor`: the two matrices equal the reduced Laplacians computed from the edge bitmasks 510 and 615;
- `Tlmc1238.K4_edge_in_two_triangles`: both triangles through the edge $\{0, 1\}$ exist and have distinct edge sets;
- `Tlmc1238.seq_228_not_logconcave`, `seq_44_logconcave_eq`, `seq_555_equalities`, `seq_336_not_logconcave_violation_arith`;
- `Tlmc1238.refute`: the conjunction of all failures of both clauses.

Because core lemmas in this toolchain pull `propext` through `Nat.testBit`, `Nat.lor` and overlapping match patterns, the development uses only kernel-reduced arithmetic (`rfl/decide`), structural `div/mod` bit tests, and disjoint pattern matches. The script `reproduce.py` (pure standard library, ~ 15 s) recomputes every number here — unimodular products, determinantal divisors, the four-vertex cactus classification, and the full six-vertex sweep.

8 Scope of the disproof

Remark 2. We refute the literal two-clause statement of Conjecture 00000001238 at the concrete graphs $K_{2,4}$, K_4 (with K_5 and $C_3 \vee C_3 \vee C_6$ as reinforcements). The Smith theory of Laplacians, the matrix-tree theorem, and the cactus decomposition of Jacobians of cycle sums are of course true and are used as tools; the log-concavity claim and its “exactly cycles” equality claim are false as stated.